

HackCelestial 3.0 — Presentation Dossier & Pitch Guide

Mahatma Education Society's Pillai University | Tech-Alegria
Project: Celeste — Local Experience Intelligence Platform

Slide 1: Identification, Focus Area & Pitch Summary

01. Identification	Team Name: BigSmokeweb Project: Celeste (Local Experience Intelligence) Repo: https://github.com/BigSmokeweb/Hackathon Live: https://experience-platform-sigma.vercel.app/
02. Problem Statement	Hyper-Local Experience Discovery, Dynamic Route Optimization & Verified Artisan Marketplace for Cultural Tourism in India.

03. Pitch Summary / Abstract (96 words):
Most travel platforms rely on commercialized, SEO-bloated listicles that trap travelers in tourist crowds while overlooking verified cultural artisans and hidden gems. Celeste solves this through a dual-engine architecture: a deterministic PostGIS spatial filtering and mathematical scoring engine paired with an isolated, privacy-compliant AI phrasing proxy. Travelers receive real-time, weather-adaptive itineraries with budget and time continuity constraints, while verified local hosts manage authentic offerings backed by Argon2 authentication, Supabase Postgres, and strict KYC verification. The result is a resilient, sub-second discovery engine empowering local tourism with zero hallucinations and complete DPDP privacy compliance.

Slide 2: Proposed Solution

Celeste is an end-to-end local experience intelligence platform and marketplace connecting travelers with verified cultural artisans and hosts across Indian heritage cities (Mumbai, Jaipur, Ahmedabad, etc.).

- **1. Deterministic 10-Factor Scoring Engine:** Discards LLM hallucination in venue selection. Uses native PostGIS geography queries (geography(Point, 4326)) for instant sub-millisecond radius search, evaluating proximity, budget, authenticity, and transit continuity.
- **2. Stateless AI Phrasing Proxy:** Decoupled Google Gemini API acts purely as a linguistic layer to synthesize personalized 'Why this matches you' explanations without database authority or latency bottlenecks.
- **3. Live Ephemeral Journey & Itinerary Planner:** Turn-by-turn routing with Leaflet/OpenStreetMap, travel duration estimation, and multi-stop budget tracking.
- **4. Host Guild Portal with Verified KYC:** Direct provider onboarding with structured availability rules and private, presigned S3/MinIO bucket storage for KYC document verification.

How it Addresses the Problem:

- **Eliminates Tourist Traps:** Experiences ranked by peer authenticity scores, not sponsored ad spend.
- **Prevents Transit Fatigue:** Route continuity algorithms prevent chaotic zig-zagging across cities.
- **Guaranteed Privacy (DPDP Act):** All user coordinates are coarsened into anonymized geohashes before logging.

Slide 3: Flow Chart & Architecture

Operational Flow (Start to Finish):

- **Step 1 (Input):** Traveler selects City, Budget Band (■ to ■■■■■), Available Time, Category Vibe & Group Size.
- **Step 2 (Spatial Filter):** Backend executes PostGIS ST_DWithin radius query against Supabase PostgreSQL.
- **Step 3 (Deterministic Scoring):** 10 weighted factors rank candidates (Authenticity 20%, Proximity 15%, Budget 15%, etc.).
- **Step 4 (AI Phrasing):** Top-N candidates passed as structured JSON to Gemini Proxy for concise rationale generation.
- **Step 5 (Interactive Route):** Client renders interactive Leaflet map, turn-by-turn route, and dynamic itinerary cards.
- **Step 6 (Session Feedback):** Swiping or rejecting an item applies instant session penalty weight and recalculates the path.

Multi-Tier System Architecture Overview:

Tier	Components & Technologies	Responsibilities
Frontend Client	Next.js 14 (App Router), React 18, Tailwind CSS, Leaflet Maps, Zustand	SSR/ISR rendering, travel journal drawer, interactive maps, zero-CLS layout
API Gateway & Auth	NestJS 10, Passport JWT, Argon2id, Redis Throttler (Token Bucket)	Rate limiting, Zod validation pipes, role-based access control, cryptographic hashing
Core Services	Recommendation Engine, Itinerary Planner, Provider Service, Gemini AI Proxy	Spatial querying, 10-factor deterministic scoring, stateless LLM reasoning, session management
Data & Storage	Supabase PostgreSQL 15, PostGIS, Redis, MinIO / S3 Private KYC Bucket	High-performance spatial indexing, session caching, secure short-lived KYC URLs

Slide 4: Innovation & Unique Functionality

- **Isolated 2-Layer Recommendation Engine:** Pure mathematical filtering guarantees deterministic candidate ranking without hallucinating venues, while Gemini generates human warmth and context without ever having database access.
- **DPDP-Compliant Privacy Architecture:** Never stores or logs raw GPS tracking history. Geolocation is converted to coarse geohash grids in all logs, honoring India's Digital Personal Data Protection Act.
- **Dynamic Weather Adaptability:** Real-time classification into INDOOR, OUTDOOR, and WEATHER_DEPENDENT dynamically boosts covered cultural spots during monsoon rains or heatwaves.
- **Session-Scoped Rejection Learning:** Rejections temporarily bias the current trip itinerary without permanently corrupting the user's permanent preferences.
- **Argon2id Enterprise Hashing:** State-of-the-art memory-hard password hashing protecting both travelers and local hosts.

Slide 5: Technical Details, Deployment & Cost

Stack Matrix: Next.js 14 App Router (Frontend) | NestJS 10 (Backend) | Supabase Postgres + PostGIS (DB) | Redis (Cache) | Google Gemini 1.5 Flash (AI Proxy).

Deployment Topology: Frontend on Vercel Edge CDN with automatic asset compression. Backend containerized with Docker on Railway/Render. Database on Supabase Tokyo region (aws-0-ap-northeast-1) utilizing connection pooling.

Estimated Monthly Operational Costs (10,000 MAU):

Component	Service / Plan	Estimated Cost (USD/mo)
Frontend Hosting	Vercel Hobby / Pro Tier (Edge CDN)	\$0 – \$20
Backend Compute	Railway / Render (1GB RAM, 1 vCPU)	\$7 – \$15
Database & PostGIS	Supabase Managed PostgreSQL (Free / Pro Tier)	\$0 – \$25
Cache & Rate Limit	Upstash Serverless Redis	\$0 – \$5
AI Reasoning Proxy	Google Gemini 1.5 Flash API (Pay-as-you-go)	\$0 – \$8
Total Operational Cost	Ultra-lean Serverless Architecture	~\$15 – \$65 / month

Slide 6: Existing Solutions & Comparison

Evaluation Dimension	Traditional Portals (TripAdvisor/MMT)	Pure AI Planners (RoamAround/Mindtrip)	Celeste Platform (Our Solution)
Discovery Engine	Sponsored ads & popularity bias	Pure LLM generation (hallucinations)	Deterministic PostGIS + AI Phrasing Proxy
Host & Artisan Vetting	Unvetted crowdsourcing	None (scraped data)	Host Guild KYC verification
Spatial Route Continuity	Static list; no transit logic	Often unfeasible transit paths	Turn-by-turn routing with distance penalty
Data Privacy	Extensive tracking cookies	Variable third-party logging	DPDP compliant (coarse geohashes)
System Resilience	Requires full connectivity	Breaks on API outage	Instant fallback to template scoring

Slide 7: Supplementary Information & Elevator Pitch

Live Production Website:	https://experience-platform-sigma.vercel.app/
GitHub Code Repository:	https://github.com/BigSmokeweb/Hackathon
Interactive Demo Video:	YouTube / Google Drive Walkthrough Video Link

Live Pitch Script (30-Second Elevator Pitch):

*'Good morning judges. Traditional travel apps push commercialized tourist traps, while modern AI trip planners hallucinate fake locations and ignore real-world geography. We built **Celeste** — an intelligent local experience platform designed for India's cultural tourism. Celeste replaces AI guesswork with a deterministic 10-factor scoring engine powered by Supabase PostGIS, paired with an isolated Gemini AI phrasing proxy for natural explanations. With Argon2-secured authentication, full DPDP privacy compliance, and a dedicated Artisan Guild portal, Celeste delivers personalized, verified, and route-optimized journeys in under 200 milliseconds. Thank you!'*